ODIR-5K Dataset

Final Exploratory Data Analysis (EDA) Report

Dataset structure • demographics • diagnostic annotations • eye-level patterns • image integrity

Project scope: Deep exploratory analysis of the ODIR-5K ocular disease recognition dataset. This report summarizes dataset properties and annotation patterns; it does not train or evaluate machine-learning models.

Analysis dataset: 6,392 selected-eye records; 26 columns in the final working DataFrame. The original CSV was previously inspected as 6,392 rows and 19 columns; later EDA added derived analysis columns.

Important interpretation note: All percentages and relationships in this report describe this dataset and its annotations. They should not be interpreted as population disease prevalence or clinical conclusions.

1. Executive Summary

The EDA covered file inventory, tabular data quality, patient demographics, disease indicator flags, encoded labels, diagnostic keyword strings, age patterns, left-right eye annotation differences, visual image inspection, and final integrity checks.

Key takeaways:

• The working dataset contains 6,392 records and 26 columns, including derived EDA columns.
• No missing values or exact duplicate rows were found in the final DataFrame.
• All 6,392 label values matched their one-hot target representations after parsing the string-encoded lists.
• All 6,392 referenced preprocessed image files were found.
• Ages 50–69 account for 4,134 records (64.68%).
• Exact left-right diagnostic keyword strings matched in 3,692 records (57.76%).
• The eye-level label distribution and indicator-flag distribution are different and must not be treated as interchangeable.

2. Dataset Overview

Item	Finding
Dataset	ODIR-5K (Ocular Disease Intelligent Recognition)
Working DataFrame shape	6,392 rows × 26 columns
Original tabular schema previously inspected	6,392 rows × 19 columns
Unique patient IDs	3,358
Rows associated with repeated patient IDs	6,068
Preprocessed images	6,392 files referenced by the working table
Preprocessed image dimensions from earlier inspection	512 × 512
Original image dimensions from sampled comparison	2,304 × 1,728
Exact duplicate rows	0
Missing values in final working DataFrame	0

The repeated patient IDs reflect multiple eye records associated with the same patient identifier. Repeated IDs are not automatically duplicate observations.

3. File Inventory and Image Inspection

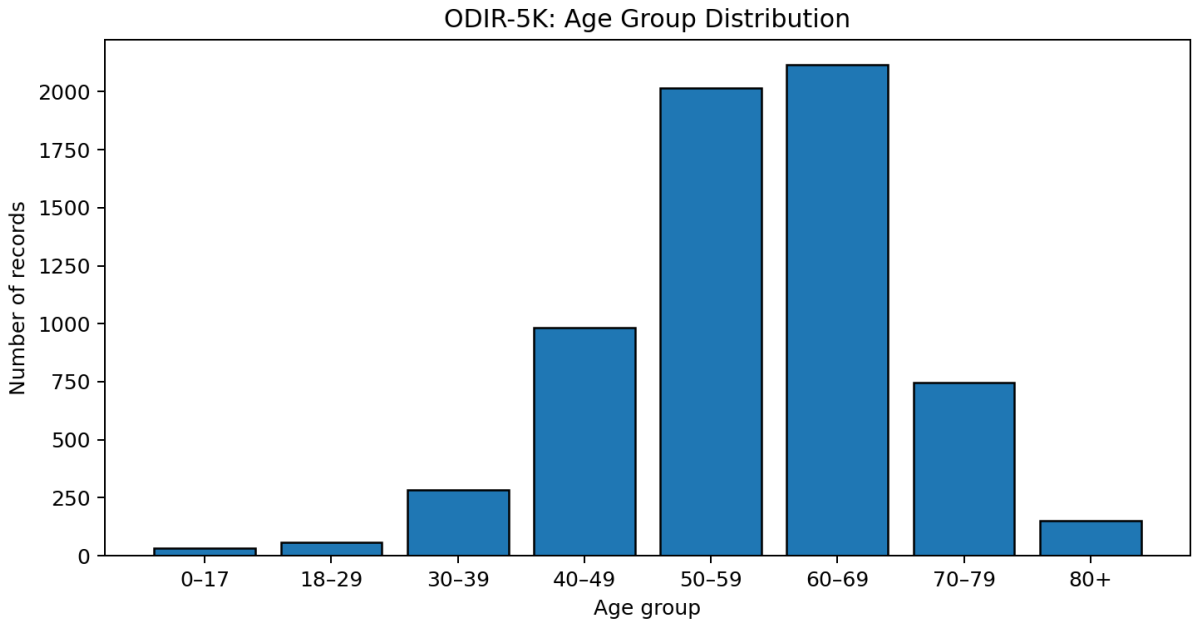
The extracted dataset inventory contained 14,394 files: 14,392 JPG images, one CSV file, and one XLSX file. The original ODIR-5K image directory contained 8,000 images, while the preprocessed image directory contained 6,392 images. Sampled visual comparison showed that the original images (2,304 × 1,728) and preprocessed images (512 × 512) differ in framing and scale. Only a small number of image pairs were visually inspected, so this is not a full clinical or image-quality audit.

Technical validation confirmed that all 6,392 image paths referenced by the working dataset exist in the preprocessed image directory. The path check confirms file presence, not clinical quality or readability for every image.

4. Demographic Analysis

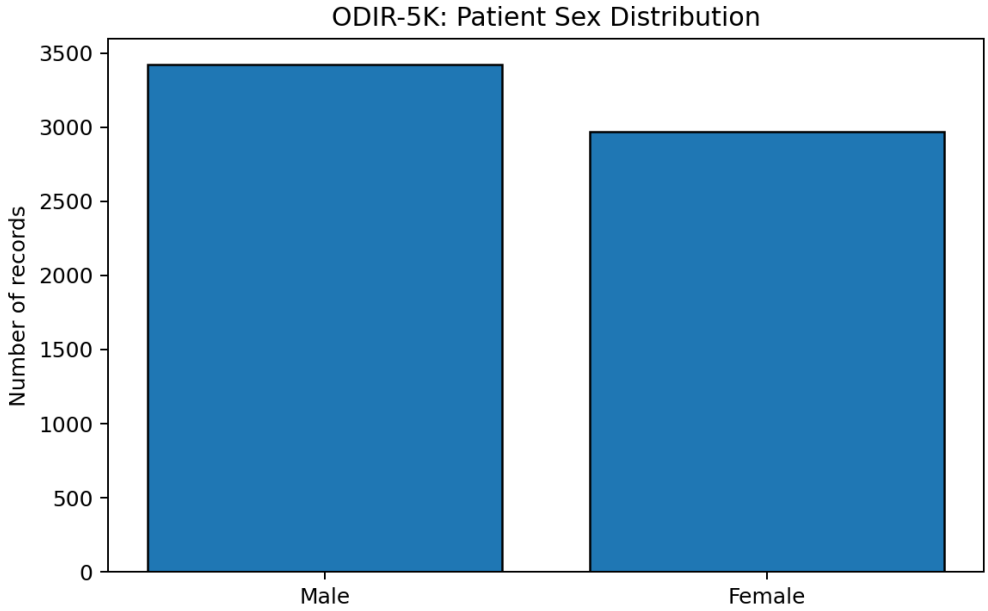
4.1 Patient age

Mean age was 57.86 years (standard deviation 11.73), with a median of 59 years and observed range of 1–91 years. The largest age groups were 60–69 (2,117 records; 33.12%) and 50–59 (2,017; 31.56%). Together, ages 50–69 account for 64.68% of records.



4.2 Patient sex

The dataset included 3,424 male records (53.57%) and 2,968 female records (46.43%). This describes the composition of the dataset and is not evidence of sex-specific disease risk.

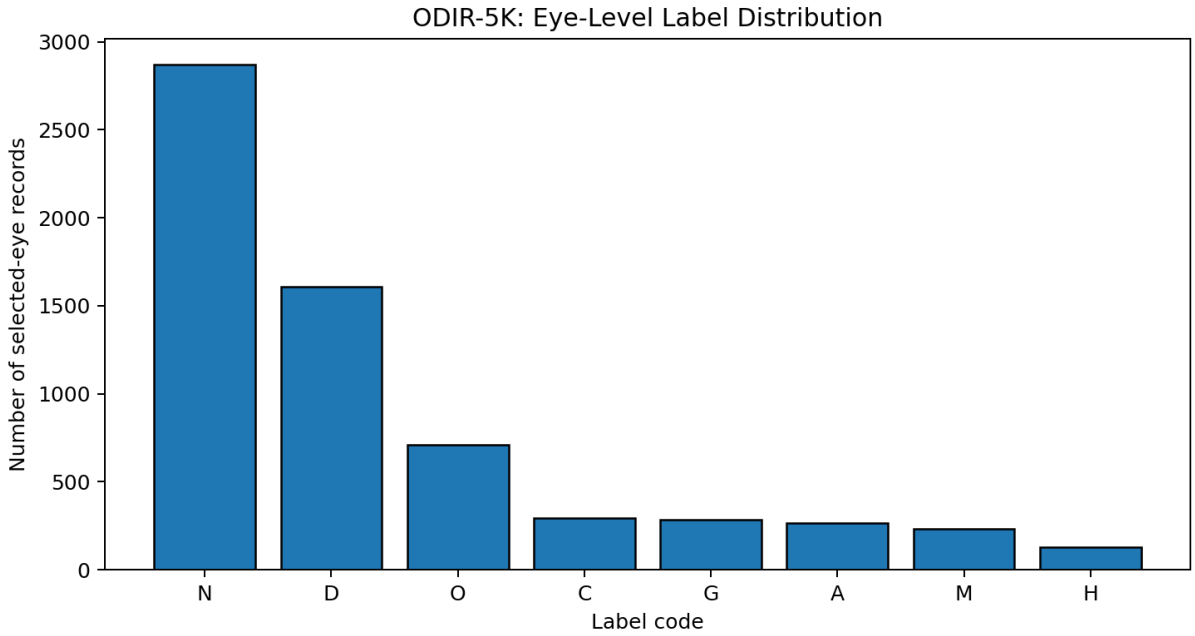


5. Diagnostic Labels and Disease Indicators

5.1 Eye-level label distribution

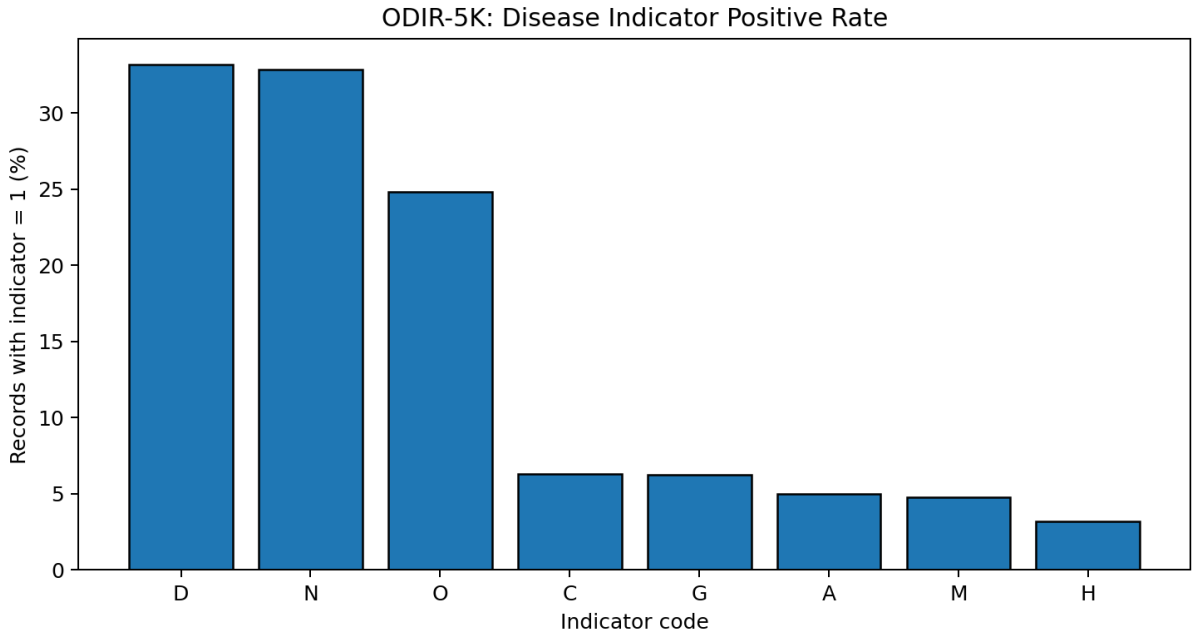
The selected-eye label distribution was: N 2,873 (44.95%), D 1,608 (25.16%), O 708 (11.08%), C 293 (4.58%), G 284 (4.44%), A 266 (4.16%), M 232 (3.63%), and H 128 (2.00%). These are eye-level records represented by the `labels`

column.


ODIR-5K: Eye-Level Label Distribution
Number of selected-eye records
3000
2500
2000
1500
1000
500
0
N
D
O
C
G
A
M
H
Label code

5.2 Disease indicator flags

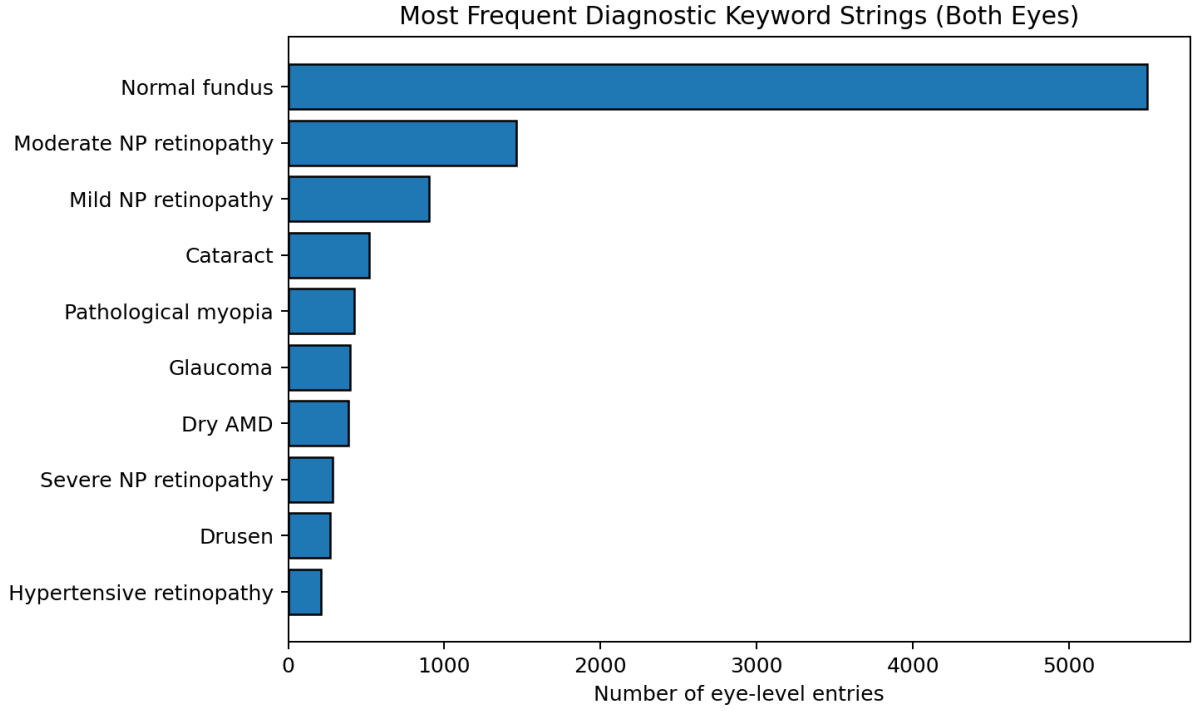
Positive indicator counts were D 2,123 (33.21%), N 2,101 (32.87%), O 1,588 (24.84%), C 402 (6.29%), G 397 (6.21%), A 319 (4.99%), M 306 (4.79%), and H 203 (3.18%). Indicator flags are not mutually exclusive: 5,391 rows had one active indicator, 955 had two, and 46 had three. In total, 1,001 records (15.66%) had multiple active indicators.



Important distinction: the indicator flags and the `labels` column are not interchangeable. Exact set agreement between the decoded labels and the active indicators was observed in 4,667 rows (73.01%); the label set was contained within the indicator set in 5,617 rows (87.88%). This discrepancy should be documented rather than silently harmonized.

6. Diagnostic Keyword Analysis

The left-eye and right-eye diagnostic keyword columns were analyzed as complete strings. Across both columns there were 12,784 eye-level entries. The most frequent combined strings were normal fundus (5,501; 43.03%), moderate non-proliferative retinopathy (1,462; 11.44%), mild non-proliferative retinopathy (900; 7.04%), cataract (518; 4.05%), pathological myopia (424; 3.32%), glaucoma (397; 3.11%), and dry age-related macular degeneration (388; 3.04%).

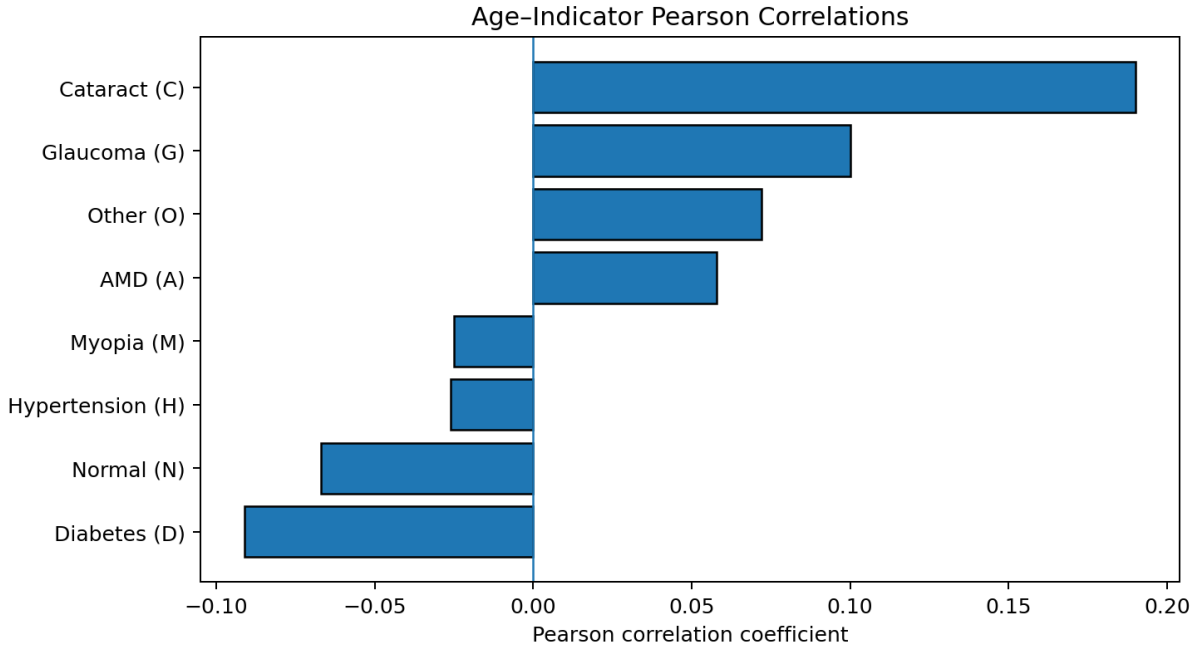


Multi-description strings were extremely rare: 12,782 entries (99.98%) contained one comma-separated description, and only two entries (0.02%) contained two descriptions. The two entries were “hypertensive retinopathy,diabetic retinopathy” and “low image quality,maculopathy.”

These keyword counts count complete strings, not every individual condition mentioned inside a string. They reflect dataset annotations rather than independently confirmed clinical diagnoses.

7. Age and Disease Indicator Relationships

Pearson correlations between age and indicator flags were generally weak. The largest positive association was with cataract (C, r = +0.190), followed by glaucoma (G, +0.100), other (O, +0.072), and AMD (A, +0.058). Normal (N, –0.067) and diabetes-related indicator D (–0.091) had weak negative correlations in this dataset. M and H were close to zero (–0.025 and –0.026).

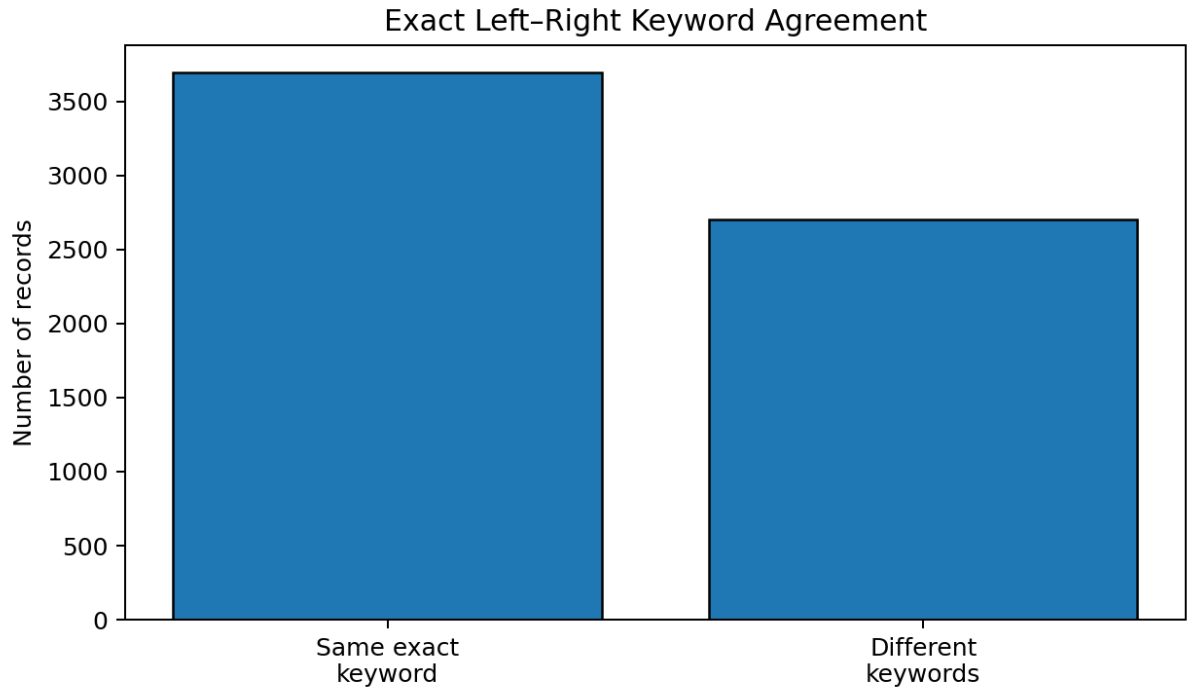


Age-group percentages also showed descriptive patterns: cataract indicators increased in older groups, reaching 38.82% among records aged 80+, while glaucoma reached 13.82% in that group. The D indicator was more common in middle-age groups than in the oldest group. The 80+ group contained 152 records, so small subgroup counts and dataset composition should be considered.

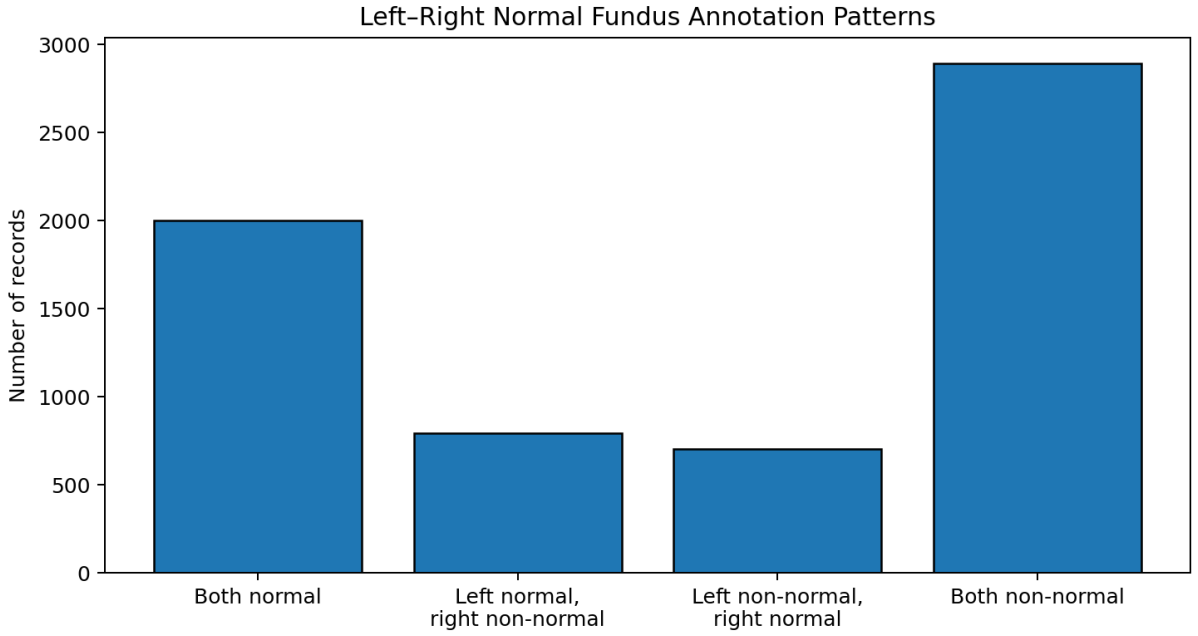
Statistical caution: small p-values were observed for several age associations, but records from the same patient may not be independent, and multiple comparisons were made. Correlation does not establish causation or population-level clinical risk.

8. Left-Eye vs Right-Eye Annotation Patterns

Exact keyword agreement was found in 3,692 of 6,392 rows (57.76%); 2,700 rows (42.24%) had different strings. The most common exact matching pair was normal fundus in both eyes (2,002 records). The most common differing pair was left normal fundus and right mild non-proliferative retinopathy (174 records); the reverse appeared 120 times.



Using the exact string “normal fundus” as the definition of normal annotation, 2,002 records (31.32%) had both eyes normal; 794 (12.42%) had left normal/right non-normal; 703 (11.00%) had left non-normal/right normal; and 2,893 (45.26%) had both eyes non-normal.



“Non-normal” here is an operational text category: any keyword other than the exact string `normal fundus`. It is not a verified disease status. Differences between eyes can reflect eye-level variation, annotation practice, or image quality.

9. Disease Indicator Co-occurrence

The N indicator did not co-occur with any other indicator in the reviewed data. Its pairwise correlations were negative with other flags, including N–D (–0.493) and N–O (–0.402). This may reflect the dataset's annotation convention for the normal class and should not be interpreted as a clinical relationship.

Other indicator relationships were weak. The highest positive pairwise correlation was D–H (+0.041). D–O had the largest observed co-occurrence count (497) among the noted pairs, despite a small negative correlation (–0.023). This illustrates that co-occurrence counts and correlation coefficients answer different questions, particularly with imbalanced or mutually exclusive flags.

10. Data Quality and Final Validation

Validation check	Result	Interpretation
Missing values	0	Passed
Exact duplicate rows	0	Passed
Unique patient IDs	3,358	Repeated IDs expected across eye records
Label-target agreement after parsing	6,392 / 6,392 (100%)	Passed
Existing preprocessed image paths	6,392 / 6,392	Passed
Missing referenced image files	0	Passed

The initial direct string comparison between `labels` and `target` produced 0 matches because they were stored in different string formats. After parsing the string-encoded lists and decoding the one-hot vector, all 6,392 records matched. The issue was representational rather than a confirmed label inconsistency.

11. Limitations

• This is an observational EDA of a curated dataset; it does not estimate disease prevalence in the general population.
• Labels, diagnostic keywords, and indicator flags may follow different annotation conventions and should not be assumed equivalent.
• Repeated patient IDs mean eye-level rows may not be statistically independent.
• Image inspection covered samples rather than every image; path existence does not establish image quality.
• Correlation and age-group comparisons are descriptive and do not establish causality.
• Complete keyword-string counts do not count individual disease mentions inside combined strings.

12. Conclusion and Recommended Next Steps

The EDA established a reproducible descriptive profile of the ODIR-5K working dataset. The table passed checks for missing values, exact duplicate rows, label-target consistency after normalization, and referenced preprocessed image existence. The dataset is concentrated in ages 50–69, includes a range of eye-level labels and indicator flags, and shows meaningful left-right annotation differences.

Before any downstream use, preserve the distinction between the original dataset columns and derived EDA columns, document the mapping and conventions for `labels`, `target`, and indicator flags, and treat patient-level grouping carefully in any future statistical analysis. This project intentionally stops at EDA and does not train a machine-learning model.

Report generated from the results recorded during the ODIR-5K EDA session on 23 October 2026.

ODIR-5K Dataset EDA

Detailed Annotation Analysis — supplemental section for the final EDA report

Purpose

This section examines how diagnostic annotations are represented in the ODIR-5K CSV, how the left- and right-eye keyword fields compare, how the human-readable labels relate to the encoded target, and why the eight disease-indicator columns must be interpreted separately. The results are descriptive dataset checks, not clinical conclusions or a model evaluation.

1. Annotation fields and their roles

Field / representation	What it contains	How to use it in EDA
Left-Diagnostic Keywords	Text annotation for the left-fundus record, such as “normal fundus”, “cataract”, or a retinopathy description.	Analyze exact strings and keyword frequencies; preserve original text before any normalization.
Right-Diagnostic Keywords	Equivalent text annotation field for the right-fundus record.	Compare with left-eye text to study annotation agreement and differences.
labels	Human-readable label representation associated with the selected record/image.	Parse into a list before counting categories; do not count raw string characters.
target	Encoded multi-class target vector.	Decode using the documented class order, then validate against parsed labels.
N, D, G, C, A, H, M, O	Eight binary indicator columns: Normal, Diabetes, Glaucoma, Cataract, AMD, Hypertension, Myopia, Other (dataset abbreviations).	Summarize each indicator independently and inspect co-occurrence. Do not assume these columns are interchangeable with labels.

The working table contains 6,392 records and 26 columns after derived EDA features were added. The source CSV has 19 columns. The selected-eye convention is balanced: 3,198 left-eye records and 3,194 right-eye records. The dataset also contains 3,358 unique patient IDs, so records are not independent patient-level observations.

2. Diagnostic keyword distribution

Combining the left- and right-eye keyword fields yields 12,784 eye-side annotation entries (6,392 records × two keyword fields). The most frequent exact keyword strings are dominated by normal-fundus and diabetic-retinopathy descriptions.

Keyword string	Count	Share of 12,784 entries
Normal fundus	5,501	43.03%
Moderate non-proliferative retinopathy	1,462	11.44%
Mild non-proliferative retinopathy	900	7.04%
Cataract	518	4.05%
Pathological myopia	424	3.32%
Glaucoma	397	3.11%
Dry AMD	388	3.04%
Severe non-proliferative retinopathy	≈286	≈2.24%
Drusen	≈270	≈2.11%
Hypertensive retinopathy	≈212	≈1.66%

Interpretation: “Normal fundus” is the largest annotation string, followed by diabetic-retinopathy severity descriptions. The long tail of less frequent strings means that a future analysis should distinguish exact-string frequency from grouped clinical concepts. Counts marked with ≈ are approximate values from the earlier EDA summary and should be checked against the live DataFrame before publication.

3. Left-eye and right-eye annotation comparison

Exact-string agreement is a strict text comparison: capitalization, spelling, punctuation, and wording differences can make two semantically similar descriptions count as different. Using the original keyword strings, left and right annotations match exactly for 3,692 of 6,392 records (57.76%); 2,700 records (42.24%) have different strings.

Comparison	Records	Share
Exact same keyword string on both sides	3,692	57.76%
Different keyword strings	2,700	42.24%
Both sides annotated “normal fundus”	2,002	31.32%
Left normal; right not normal	794	12.42%
Left not normal; right normal	703	11.00%
Neither side annotated “normal fundus”	2,893	45.26%

The most common matching pair is “normal fundus” on both sides (2,002 pairs), followed by matching moderate non-proliferative retinopathy (452), cataract (158), dry AMD (158), and glaucoma (154). Common mismatches include left normal fundus/right mild non-proliferative retinopathy (174) and the reverse pattern (120).

Important interpretation

The “normal” vs. “not normal” split above is operationally based on whether the exact annotation string is “normal fundus”. It does not independently confirm disease status, severity, or image quality. Different annotations between eyes may be clinically plausible, but this EDA alone cannot validate the clinical correctness of either annotation.

4. Label and target consistency

A direct comparison of the raw labels and target strings is not meaningful because they use different representations. The EDA parsed the labels into a list and decoded the target vector using the class order [N, D, G, C, A, H, M, O]. After this normalization, all 6,392 records matched: 6,392/6,392 (100%), with zero mismatches.

Validation check	Result	Meaning
Labels parsed successfully	6,392 / 6,392	All label entries were interpretable by the parsing procedure.
Decoded target agrees with labels	6,392 / 6,392	The two representations are consistent after decoding.
Mismatch count	0	No label-target inconsistencies were detected in this check.

This is a useful data-integrity result, but it only establishes internal consistency between two fields. It does not prove that the underlying medical annotation is correct.

5. Diagnostic labels versus disease indicators

The human-readable label representation and the eight binary indicator columns should not be treated as interchangeable. Exact set agreement between labels and indicator-derived categories was 4,667 records (73.01%). The label categories were contained within the indicator set for 5,617 records (87.88%). This indicates meaningful disagreement between the two representations.

Check	Records	Share
Exact set agreement	4,667	73.01%
Labels contained within indicator-derived set	5,617	87.88%
Not exact set agreement	1,725	26.99%

Indicator prevalence in the 6,392-row working table:

Indicator	Positive rows	Positive rate
D	2,123	33.21%
N	2,101	32.87%
O	1,588	24.84%
C	402	6.29%
G	397	6.21%
A	319	4.99%
M	306	4.79%
H	203	3.18%

Indicator columns are not mutually exclusive: 5,391 records have one active indicator, 955 have two, and 46 have three. Thus, 1,001 records (15.66%) have multiple active indicators. In this dataset, N never co-occurs with any of the other seven indicators; this produces negative correlations between N and other indicators. Treat this as an observed dataset/annotation pattern, not as a general clinical rule.

6. Annotation structure and edge cases

Across the two keyword fields, there are 12,784 annotation entries. Nearly all entries are single-description strings: 12,782 (99.98%) contain one description, while only two entries (0.02%) contain multiple comma-separated descriptions:

Multi-description entry	Count
hypertensive retinopathy, diabetic retinopathy	1
low image quality, maculopathy	1

This matters for preprocessing. If multi-description strings are present, splitting on commas may be useful for concept-level analysis, but it can also accidentally split legitimate phrases if the delimiter is not standardized. Keep the raw annotation fields and create a separate normalized field for any grouped analysis.

7. Recommended annotation-analysis workflow

Step	Recommended practice
1. Preserve	Keep raw left/right keyword strings unchanged as the source of truth.
2. Normalize	Create lowercased, whitespace-trimmed copies for robust text comparisons; document any synonym mapping.
3. Count	Report both exact-string counts and concept-level grouped counts, clearly distinguishing them.

Step	Recommended practice
4. Compare eyes	Use exact agreement and normalized/concept agreement separately; keep left/right asymmetry visible.
5. Validate labels	Parse labels and decode target vectors with a fixed, documented class order.
6. Audit indicators	Compare indicator-derived categories with labels and investigate disagreements rather than silently forcing them to match.
7. Sample-review	Manually inspect a stratified sample of image–annotation pairs with domain expertise before making clinical claims.

8. Limitations and conclusions

This annotation analysis is descriptive. Exact text matching can undercount semantic agreement; keyword frequencies are affected by repeated patients and two-eye records; and the CSV's text labels, encoded targets, and binary indicators follow different conventions. Some records are linked to the same patient ID, so statistical tests treating every row as independent may overstate certainty. The findings do not establish prevalence in the general population or diagnostic performance.

Conclusion: ODIR-5K's annotation fields are internally consistent between parsed labels and decoded targets in this check, but the free-text keyword fields and binary disease indicators do not fully agree. A rigorous downstream workflow should preserve raw annotations, document label mappings, quantify left/right differences, and manually review representative image–annotation pairs before using the data for clinical interpretation.

ODIR-5K EDA — Classification Framework

Problem definition, independent features (X), dependent target (y), and classification type

1. Can ODIR-5K be used for classification?

Yes. ODIR-5K can support supervised classification. The exact task depends on how the target is defined and whether each image is assigned one class or can have multiple labels. For this report, the proposed starting task is **fundus-image multi-class classification**, provided that the validated image-level target contains exactly one class per selected image.

2. Independent features (X) versus dependent variable (y)

Role	ODIR-5K field / input	Explanation
Independent input (X)	Fundus image pixels (Left-Fundus or Right-Fundus, using the selected image path)	Primary input for an image-classification task. The model learns visual patterns from the retinal photograph.
Optional tabular inputs	Patient Age, Patient Sex	May be included in a separate or multimodal experiment if permitted by the project design. These are not image pixels and should be documented separately.
Dependent target (y)	`labels` or decoded `target`	The ground-truth class assigned to the selected image/record. Choose one representation as the target after validation; do not use both as separate predictors.
Alternative target setup	N, D, G, C, A, H, M, O	Can be modeled as separate binary outcomes or a multi-label target only if the intended task and annotation semantics justify it. These indicators should not automatically replace `labels` because EDA found disagreements.
Identifiers / metadata	ID, filename, filepath	Use for record linkage and splitting, not as predictive features. Including identifiers can cause leakage or spurious learning.

3. Why the proposed task is multi-class classification

Multi-class classification predicts exactly one class from three or more mutually exclusive classes. If each selected fundus image has one validated class among Normal (N), Diabetes/Diabetic Retinopathy (D), Glaucoma (G), Cataract (C), Age-related Macular Degeneration (A), Hypertension (H), Myopia (M), or Other (O), then the target y is a single class label. The precise class meanings and mapping must follow the dataset's documented label convention.

Component	Example
Input X	One selected retinal fundus image
Target y	One validated class, e.g. D
Model output	Predicted class plus class probabilities
Evaluation	Confusion matrix, per-class precision/recall/F1, macro-F1, balanced accuracy; report accuracy as a secondary metric if classes are imbalanced.

4. Multi-class vs. multi-label: do not confuse them

Task type	Target format	When appropriate
Binary classification	One of two classes, e.g. Normal / Abnormal	Only if the task is explicitly reduced to two validated categories.
Multi-class classification	Exactly one class per image, e.g. one of N/D/G/C/A/H/M/O	Use when the chosen image-level annotation assigns one mutually exclusive class.
Multi-label classification	A set/vector of labels; more than one label may be positive	Use only if the annotation definition permits multiple simultaneous labels for the same image.

5. Important data-leakage and validation precautions

• Use a patient-level train/validation/test split so both eyes from the same patient cannot appear across different splits. The EDA found 3,358 unique patient IDs and repeated records per ID, so a row-wise random split could leak patient-specific information.

• Keep target columns out of X. Do not include `labels`, `target`, disease indicator columns, or diagnostic keyword fields as predictors when those fields define the target. That would let the model see the answer.

• Do not use ID, filename, or filepath as predictive features. They are identifiers, not clinically meaningful image features.

• If the project is image classification, use the fundus image itself as X. Age and sex can be optional metadata in a separately documented experiment; a model using only those fields is tabular classification, not image classification.

• Before training, verify that every selected image maps to exactly one target class. If the annotation is genuinely multi-label, reformulate the task as multi-label classification rather than forcing a single class.

6. Recommended project definition

Task: Supervised fundus-image classification on ODIR-5K.
Independent input (X): Selected retinal fundus image pixels.
Dependent variable (y): Validated image-level class from the parsed `labels` / decoded `target` representation.
Proposed type: Multi-class classification only after confirming one mutually exclusive class per image.
Out of scope for this EDA report: Model training, tuning, and performance claims.

This is a proposed modeling formulation, not a claim that a model has been trained. The EDA validates data representations but does not establish clinical diagnostic accuracy.